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Identification and validation of key predictive factors for heart attack diagnosis using machine learning and fuzzy clustering

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ABSTRACT

Heart disease remains a leading cause of mortality worldwide, making early diagnosis and prevention critical for improving patient outcomes. However, accurate diagnosis often depends on physician expertise, which can lead to medical errors. This study develops a machine learning-based predictive model to assist physicians in diagnosing and predicting heart attacks with higher precision. Patient data from an Iranian hospital were collected and preprocessed, including handling missing values, normalization, and outlier removal. Analysis using RapidMiner software identified seven significant predictors of heart attack from thirteen initial factors, achieving a high accuracy of 97.86%, confirmed by hospital physicians. Among various machine learning algorithms, the decision tree demonstrated the highest accuracy (97%) and provided interpretable decision pathways for clinical use. Decision rules extracted from the decision tree highlighted chest pain, trait-anxiety, age, smoking, and resting electrocardiographic results as the most influential factors. To validate these findings, fuzzy clustering was employed, revealing an alignment rate of 76% for the two most critical factors—chest pain and trait-anxiety. These findings emphasize the prioritization of these factors during initial evaluations, followed by age, smoking, and resting electrocardiographic results to enhance diagnostic accuracy. The integration of machine learning and fuzzy clustering offers a robust approach to clinical decision-making, verified by expert feedback from hospital physicians, and contributes to reducing diagnostic errors.

1. Introduction

This research investigates and analyzes methods for identifying and predicting heart attacks in patients, a critical area given that heart disease is currently the leading cause of mortality worldwide. Heart disease continues to be a global public health crisis, with its increasing prevalence attributed to modern lifestyle factors. This research is grounded in the theoretical framework of health informatics and predictive analytics, which emphasizes the integration of patient-level data with advanced computational methods to improve clinical decision-making. Heart illnesses stand out as major contributors to global mortality in the modern era. One person succumbs to death from heart attacks approximately every 33 s (Regin et al., 2023). Timely identification of heart issues proves highly efficacious in the treatment of individuals with heart diseases (Ghasemieh et al., 2023). In the most recent publication from the World Health Organization (WHO), it is highlighted that heart disease is on the rise worldwide, attributed to various factors linked to contemporary lifestyles. The global annual mortality rate from cardiac arrest exceeds 18 million, as reported (Rathee et al., 2023). In 2008 alone, heart disease claimed the lives of 17.3 million individuals. Shockingly, over 80% of all deaths worldwide stem from heart disease (Dangare and Apte, 2012). The Global Burden of Diseases, Injuries, and Risk Factors Study received approval from the Institutional Review Board Committee at the University of Washington. Fig. 1 displays the number of mortalities caused by heart attacks, showing a consistent rise, starting at 12.1 million (the sum of female and male) in 1990 and escalating to 18.6 million by 2019 (Gregory et al., 2020).

Heart disease stands as the leading cause of death among males and females in the United States. Only in this century were heart attacks acknowledged as a significant public health concern (Brown and Goldstein, 2020). In evaluating the clinical applicability of screening for a heart disease marker, physicians should take into account the presence of a standardized and replicable assay for the specific marker of interest. Additionally, one should consider the existence of a consistent body of

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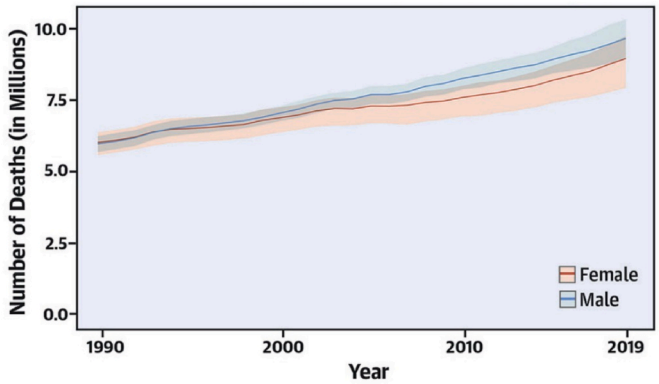

Fig. 1. The number of heart attack mortalities by sex from 1990 to 2019 (Ghasemieh et al., 2023).

prospective epidemiological studies that demonstrate the ability of elevated baseline levels of the new marker to predict future risk (Ridker and MPH, 1999). While numerous studies have explored diverse prediction algorithms, choosing the most effective predictor remains a challenging decision (Arjmand et al., 2020). Beyond crafting highly efficient predictors, certain factors also play a crucial role in achieving precise predictions (Arjmand et al., 2019). Advancements in machine learning techniques are swiftly progressing within the medical domain, enabling prompt disease detection. The application of these sophisticated methods for diagnosis offers a significant time-saving advantage (Okay and Çalışkan, 2023).

Conventional medical history has faced criticism for its lack of reliability in diagnosing heart disease. In contrast, machine learning techniques prove to be effective and dependable in distinguishing between healthy individuals and those at risk of heart attacks based on risk factors (Oliullah et al., 2023). The recent progress in applying machine learning showcases the feasibility of early-stage heart disease detection by utilizing electrocardiogram data and patient information (Ahsan and Siddique, 2022). The knowledge derived from these discoveries can be utilized by healthcare administrators to enhance the quality of service provided. Medical practitioners can also leverage this knowledge to minimize the occurrence of adverse drug effects and recommend cost-effective yet therapeutically equivalent alternatives. An important application of these techniques in healthcare management is the ability to anticipate a patient's future behavior based on their medical history (Srinivas et al., 2010).

Identifying heart disease can pose challenges due to various factors such as cholesterol, high diabetes, hypertension, and others. Evaluating the severity of heart disease in individuals necessitates the utilization of diverse techniques like data extraction and linear regression. Various methods including Decision Tree, Logistic Regression, and K-Nearest Neighbor Algorithms are employed to gauge the effectiveness of treatment. Considering the intricate nature of cardiac disease, it is imperative to approach its treatment with great care (Kathare and Gaikwad, 2021).

Despite advancements in treatment and prevention, the high prevalence and complexity of heart disease continue to strain healthcare systems globally. Early detection and risk prediction are critical to reducing the burden of heart attacks, which represent a significant portion of these cases. This research focuses on leveraging machine learning algorithms to identify and analyze key factors that contribute to heart attack risk. Traditional methods for heart attack prediction rely heavily on physician expertise and conventional risk factor models, which, while effective, may lack the precision and scalability of data-driven approaches. Given the significance of the topic, it is necessary to use tools that assist physicians in decision-making to identify the most effective factors on heart attack based on past data. This approach ensures a more accurate prediction of its occurrence.

The primary objective of this research is to identify key factors that contribute to the risk of heart attack and evaluate the predictive capabilities of machine learning algorithms using hospital data. This study aims to utilize machine learning algorithms to address the following research questions:

1. What are the key factors in predicting heart attacks?
2. Which machine learning algorithm performs best for heart attack prediction?
3. What diagnostic rules can be established from the optimal algorithm to support physicians in decision-making?

To address this question, the study aims to identify significant predictors of heart attack based on a review of literature and expert consultation, evaluate the performance of various machine learning algorithms in predicting heart attack occurrences, validate the predictive results using fuzzy clustering to assess the accuracy and relevance of the identified factors, and provide practical recommendations for healthcare professionals to improve heart attack risk management. The study uses a dataset collected from an Iranian hospital between 2018 and 2022. By focusing on this dataset, the study also offers insights into how machine learning can be applied to diverse healthcare settings. The novelty of this research lies in its approach to extracting predictive rules that help physicians make more accurate and efficient diagnoses, thereby reducing the occurrence of heart attacks. Additionally, fuzzy clustering is applied to validate the selected factors, making this approach innovative in the context of heart attack prediction.

The remaining sections are organized as follows:

- Section 2 provides a literature review.
- Section 3 details the methodology, including the dataset, variables, and range of values.
- Section 4 analyzes the data, compares predictive algorithms, and presents fuzzy clustering results.
- Section 5 concludes the study and suggests future research directions.

By proposing a novel combination of machine learning algorithms and fuzzy clustering, this study offers a valuable contribution to the field, potentially aiding healthcare professionals in achieving more accurate heart attack predictions and ultimately improving public health outcomes.

2. Literature review

Exploring the utilization of machine learning in heart disease has the potential to yield valuable findings on the diagnosis and prediction of heart attack. With this objective in mind, this section will analyze research papers that explore the utilization of machine learning algorithms in the field of heart disease. The literature review can be divided into four categories.

2.1. One classification algorithm

Ozcan and Peker (2023) utilized the Classification and Regression Tree (CART) algorithm to make predictions about heart disease and uncover decision rules that explain the connections between input and output variables. The model's reliability was confirmed by achieving an 87% accuracy rate across all performance measures. Ilayaraja and Meyyappan (2015) developed mining association rule to utilize frequent item sets for forecasting the risk level of patients with heart disease. The research employed a dataset of diverse heart disease patients, and the outcomes demonstrated that the proposed approach efficiently identifies the patients' risk level from frequent item sets. Sapra et al. (2023) adopted the dual approach: Firstly, a deep neural network was employed to diagnose coronary artery disease, achieving a remarkable predictive accuracy of 96.2% and a low error rate of 3.8%. Secondly, to address

overfitting, Gaussian noise was introduced into the model for improved performance. In the second phase, the severity of the disease was evaluated using a case-based reasoning approach. Das et al. (2009) developed a proposed system that centers around a neural networks ensemble method. experiments were conducted using this tool and achieved a classification accuracy of 89.01%.

2.2. Comparison of various classification algorithms

Jindal et al. (2021) identified which patients are at a higher risk of experiencing heart disease due to several medical factors. A system for predicting heart disease is designed to utilize a patient's medical history in assessing the probability of being diagnosed with heart disease. Various algorithms of machine learning, such as logistic regression and k-nearest neighbors, were utilized to classify and predict patients with heart disease. Aravinda et al. (2022) for prediction of heart disease used multiple algorithms, including decision tree, random forest, and neural network to analyze patients' data. The experimental findings exhibited a promising accuracy rate of around 90% in predicting heart disease. Mamun Ali et al. (Mamun et al., 2021) employed several algorithms, such as random forests, decision tree, and k-nearest neighbor on a heart disease dataset obtained from Kaggle, which resulted in a three-classification system. According to the study, the random forest method demonstrated perfect accuracy, sensitivity, and specificity, scoring 100% in each of these metrics. Pires et al. (2020) proposed various techniques to examine data on heart disease, such as neural networks, decision tree, k-nearest neighbor, combined nomenclature rule inducer, support vector machines, and stochastic gradient descent. The study indicated that the decision tree and support vector machines methods obtained the most precise results, achieving 87.69% accuracy rates. Agrawal et al. (2022) employed three distinct machine-learning classification methods to differentiate healthy individuals from those with myocardial infarction. The techniques used in this research were neural networks, support vector machines, and decision tree. The outcomes obtained from these approaches exhibited the classification metrics encompassing specificity, sensitivity, and accuracy. Analysis of the machine-learning results indicated that decision tree was the most efficient predictor for classification accuracy, followed by support vector machines and neural networks. Obasi and Shafiq (2019) proposed a solution that employs algorithms such as Logistic Regression, Bayesian, and Random Forest to develop a system to assist physicians in decision-making. The system can identify and predict occurrences of heart attack by analyzing the risk factors linked with heart disease in humans or individuals. The study's dataset includes a dataset consisting of 18 factors and 1990 instances, which underwent preparation of data before being partitioned into training sets comprising 80% of the data and test sets comprising 20% of the data. The comparison of the results indicated that the system's accuracy and performance are satisfactory, with Random Forest achieving a classification accuracy of 92.44% for heart disease. In comparison, Naive Bayes Classifier and Logistic Regression attained accuracies of 61.96% and 59.7%, respectively. Patel et al. (2021) examined the efficacy of machine learning algorithms such as Artificial Neural Networks, Naive Bayes, Decision Trees, and Support Vector Machines. To facilitate comparisons, performance metrics like precision, accuracy, F-score, and recall were utilized. These methods were utilized for identifying the most effective approach and the scenarios under which it would perform best, based on experimental studies. Kumar et al. (2018) utilized various algorithms, such as Decision Trees, J48, Random Forest, and K-Nearest Neighbors, for the prediction of heart disease to analyze heart disease in a predictive manner and identify the most effective and efficient algorithms. Masethe (Masethe and Masethe, 2014) utilized data mining techniques including J48, REPTREE, CART, and Naive Bayes to forecast occurrences of heart attack. The outcome of the study indicated a predicted accuracy rate of 99%. Kurt et al. (2008) conducted a study that compares different classification techniques for predicting the presence of coronary artery disease (CAD). The study involved a retrospective analysis of 1245 subjects, out of which 865 had CAD and 380 did not. The researchers evaluated and compared the performance of multiple algorithms to determine the optimal choice for prediction. Shah et al. (2020) discussed several attributes associated with heart disease and built a model employing classification algorithms, including Naive Bayes, K-nearest neighbor, random forest, and decision tree. The objective of their study was to predict the occurrence of heart disease in patients. According to the findings, K-nearest neighbor algorithm yielded the highest accuracy score.

2.3. Presentation of clustering algorithms

Chitra and Seenivasagam (2013) developed a tool to assist physicians in diagnosing disease more efficiently. The effectiveness of the classifier using records from 270 patients was evaluated, resulting in a classification accuracy of 92%. The results indicated that their suggested clustering algorithm can better predict the possibility of heart attack compared to other widely recognized algorithms. Sundar et al. (Sundar V et al., 2012) analyzed algorithms related to data clustering. Their focus was on examining existing methods for predicting heart disease and developing a more efficient and effective approach for this purpose. Some techniques used for heart disease prediction include naive bayes, neural network, and decision tree. k-means clustering techniques have also been utilized. By employing measures of compactness and connectedness, it has been determined that the proposed method for predicting heart disease outperforms the other three techniques. This conclusion was derived through the implementation of a software prototype. Shinde et al. (2015) introduced a system for predicting heart disease by integrating the naive Bayes and k-means clustering algorithms. This system combines the strengths of both algorithms. k-means clustering is employed to enhance the efficiency of the output by grouping various attributes. Additionally, the naive Bayes algorithm was utilized for prediction purposes.

2.4. Application of the fuzzy method in diagnosing heart disease

Krishnaiah et al. (2014) aimed to categorize patients by utilizing features gathered from the source of medicine. Fuzzy k-nearest neighbor was utilized for classifying data that are categorized into separate classes. It was found that this technique performed better than other classifiers of parametric techniques. Pathak and Valan (2019) developed a heart disease diagnostic model utilizing a rule-based approach with fuzzy logic in conjunction using a decision tree-based approach. The suggested model, employed only eight patient attributes, achieved an accuracy of 88%, indicating its statistical significance in accurately diagnosing patients with heart disease.

Numerous reviewed studies have employed diverse machine learning techniques for heart disease prediction. A comprehensive summary of previous studies in this domain has been provided in Table 1.

The novelty of the current research, based on an analysis and comparison of earlier investigations, reveals several critical research gaps that persist:

1. Feature Selection and Expert Opinion Bias: A common issue in most studies is the over-reliance on expert opinions for selecting influential features. In machine learning, feature selection should be based on data-driven methods, such as recursive feature elimination or mutual information, to reduce biases inherent in expert judgment.
2. Interpretability of Machine Learning Models: Many studies have focused on improving the accuracy of predictive algorithms, but they often neglect the importance of interpretability, which is crucial in healthcare settings. Developing models that can provide transparent decision-making pathways ("if … then" rules) would greatly enhance their clinical usability.

Table 1
Literature review comparison.

Article	Year	Country	Objective	Data set	Selection of Key Heart Attack Factors (In addition to Expert Opinions)	Classification Technique	Clustering Technique	Comparison of Various Algorithms	Application of the Fuzzy Method	Extracting Decision Rules	validating the Extracted Rules
Ozcan and Peker (Ozcan and Peker, 2023)	2023	Turkey	For heart disease modeling and prediction	IEEE DataPort		✓				✓	
Ilayaraja and Meyyappan (Ilayaraja and Meyyappan, 2015)	2015	India	To predict the risk of heart diseases through frequent itemsets	The dataset of various heart disease patients		✓					
Sapra et al. (Sapra et al., 2023)	2023	India	To detect severity of coronary artery disease	The data of Medical College Shimla		✓					
Das et al. (Das et al., 2009)	2009	Turkey	Effective diagnosis of heart disease	Cleveland heart disease database		✓					
Jindal et al. (Jindal et al., 2021)	2020	India	For heart disease prediction	UCI repository dataset		✓		✓			
Aravinda et al. (Aravinda et al., 2022)	2022	India	For the prediction of heart attacks	data on the cohort study		✓		✓			
Mamun Ali et al. (Mamun et al., 2021)	2021	Bangladesh	For heart disease prediction	Kaggle		✓		✓			
Pires et al. (Pires et al., 2020)	2020	Portugal	For the evaluation of the presence of heart disease	UCI Machine learning repository		✓		✓			
Agrawal et al. (Agrawal et al., 2022)	2022	United States of America	For classifying healthy and infarcted patients	PTB Diagnostic ECG Database		✓		✓			
Obasi and Shafiq (Obasi and Shafiq, 2019)	2019	Canada	For detecting and predicting heart attack and diseases	UCI machine learning repository		✓		✓			
Patel et al. (Patel et al., 2021)	2021	India	For heart disease prediction	Kaggle		✓		✓			
Kumar et al. (Kumar et al., 2018)	2018	India	For prediction of heart diseases	UCI machine learning repository		✓		✓			
Masethe (Masethe and Masethe, 2014)	2014	South Africa	For prediction of heart disease	medical practitioners in South Africa		✓		✓			
Kurt et al. (Kurt et al., 2008)	2008	Turkey	For predicting coronary artery disease	Cardiology Clinic of Trakya University Medical Faculty in Turkey		✓		✓			
Shah et al. (Shah et al., 2020)	2020	India	For heart disease prediction	UCI machine learning repository		✓		✓			
Chitra and Seenivasagam (Chitra and Seenivasagam, 2013)	2013	India	For heart attack prediction	UCI machine learning repository			✓	✓			
Sundar et al. (Sundar V et al., 2012)	2012	India	For predicting heart	UCI machine learning repository			✓	✓			
Shinde et al. (Shinde et al., 2015)	2015	India	For heart disease prediction	UCI machine learning repository	✓	✓	✓	✓			
Krishnaiah et al. (Krishnaiah et al., 2014)	2014	India	For diagnosis of heart disease patients	UCI machine learning repository			✓	✓	✓		
Pathak and Valan (Pathak and Valan, 2019)	2019	India	For heart disease diagnosis	UCI machine learning repository		✓	✓	✓	✓	✓	
Current Research		**Iran**	For **identification and analysis of key factors in diagnosing and predicting heart attack in patients**	**Data were collected from a hospital in Iran between 2018 and 2022**	✓	✓	✓	✓	✓	✓	✓

Table 2
Independent variables.

Row	Independent Variable	Value Range
1	Age	Continuous variable
2	Chest Pain Type	Four-category variable (number 0: no chest pain – number 1: mild chest pain – number 2: severe chest pain – number 3: unbearable chest pain)
3	Gender	Two-category variable (Female: 0 – Male: 1)
4	Weight_Category	Three-category variable (less than typical: 0 - typical: 1 - greater than typical: 2)
5	Cholesterol	Continuous variable
6	Resting-electrocardiographic-results	Two-category variable (unhealthy: 0 – healthy: 1)
7	Trait_Anxiety	Continuous variable (between 0 and 100)
8	Smoking	Two-category variable (non-smoking: 0 – smoking: 1(
9	Exercise	Three-category variable (does not exercise: 0 – does not exercise regularly: 1 – is an athlete: 2)
10	Lifestyle	Two-category variable (unhealthy: 0 – healthy: 1)
11	Hypertension	Two-category variable (unhealthy: 0 – healthy: 1)
12	Diabetes	Two-category variable (unhealthy: 0 – healthy: 1)
13	Renal Failure	Two-category variable (unhealthy: 0 – healthy: 1)

Table 3
Dependent variable.

Row	Dependent Variable	Value Range
1	Heart attack	Two-category variable (did not occur: 0- occurred: 1)

3. Integration of Clustering and Hybrid Models: Many studies have explored individual machine learning algorithms, but there is limited research on combining clustering with other models, or using hybrid approaches that integrate different machine learning techniques to improve prediction and decision-making accuracy.

This research introduces a novel method for extracting predictive rules to assist physicians in making more accurate and efficient diagnoses, ultimately reducing the occurrence of heart attacks. The approach also incorporates fuzzy clustering to validate the chosen factors, adding innovation to heart attack prediction.

3. Methodology

This study analyzed data from heart patients treated at a hospital in Iran, between 2018 and 2022. The data for this study were sourced from the hospital's Hospital Information Systems (HIS), with the collaboration of hospital IT and data management teams to facilitate the access and extraction of the necessary data. The inclusion and exclusion criteria for patient selection were carefully defined to ensure data integrity. Specifically, only patients with complete records were included in the study, while those with missing or conflicting data were excluded. This approach helped maintain the reliability of the analysis. Additionally, ethical considerations were central to the data collection process. All patient data were anonymized to ensure confidentiality, and any necessary permissions or consent for data use were obtained in accordance with ethical standards for research. After preprocessing the data, information from 135 patients was included in the analysis. The high prediction accuracy achieved using machine learning algorithms further supports the adequacy of the dataset for the study's objectives. The data were then analyzed using RapidMiner software, with the collaboration of healthcare professionals and data experts within the hospital to verify the relevance and accuracy of the extracted data.

The study, utilizing historical patient data collected from the hospital's information system, is deemed appropriate for several reasons:

- Availability of Historical Data: The hospital's data repository provided a rich dataset of patient records spanning several years, enabling the identification of meaningful patterns and associations related to heart attack risk factors.
- Efficiency in Data Analysis: This study enables researchers to utilize existing data, minimizing the time and resources needed for data collection while preserving the integrity of the analysis.
- Relevance to Research Objectives: By focusing on real-world patient records, this design aligns with the objective of identifying and validating significant predictors of heart attack risk, providing practical insights for healthcare applications.

These considerations highlight the suitability of this design for addressing the study's objectives effectively.

After consulting with specialized hospital physicians and reviewing relevant literature, the key factors impacting the risk of heart attack are as follows:

- Table 2 displays the independent variables (predictive factors), which are variables that explain and predict dependent variables.
- Table 3 displays the dependent variable, which is affected by the independent variables and changes as a result. The analysis of the problem aims to explain or predict the variability of the dependent variable.

4. Analysis

The Analysis section details the process and results of identifying the most significant factors affecting heart attacks and evaluating the predictive performance of various machine learning algorithms. This section includes the following key components:

4.1. Selection of the most important factors affecting the heart attack

The Optimize Selection module in Rapid Miner software was employed to choose the significant attributes for predicting heart attack from the 13 independent variables that were identified. This is depicted in Fig. 2 presented below:

According to Table 3 it is evident that Rapid Miner software selected 7 out of the 13 independent variables, which were later approved by hospital physicians. Using these variables listed in Table 1, the software achieved an accuracy of 97.86% in predicting heart attack, a value deemed acceptable.

4.2. Machine learning algorithms and predictive accuracy

In this phase, the data were partitioned into two sets: training (consisting of 100 patients) and testing (comprising 38 patients). The column that denoted the occurrence of a heart attack in the testing data was then excluded. The objective was to use the training data to forecast heart attacks in the testing data. Finally, the predictive accuracy of each machine learning algorithm was assessed by contrasting the predicted data in the testing set with their actual counterparts. Each algorithm was chosen based on its suitability for the research objectives of predicting heart attack risk based on various patient factors. Below, we provide an explanation for the inclusion of each algorithm:

- Decision Tree Algorithm: The decision tree was selected due to its ability to clearly identify and model the relationships between the independent variables (e.g., chest pain type, age, smoking) and the likelihood of a heart attack. It is particularly well-suited for this study

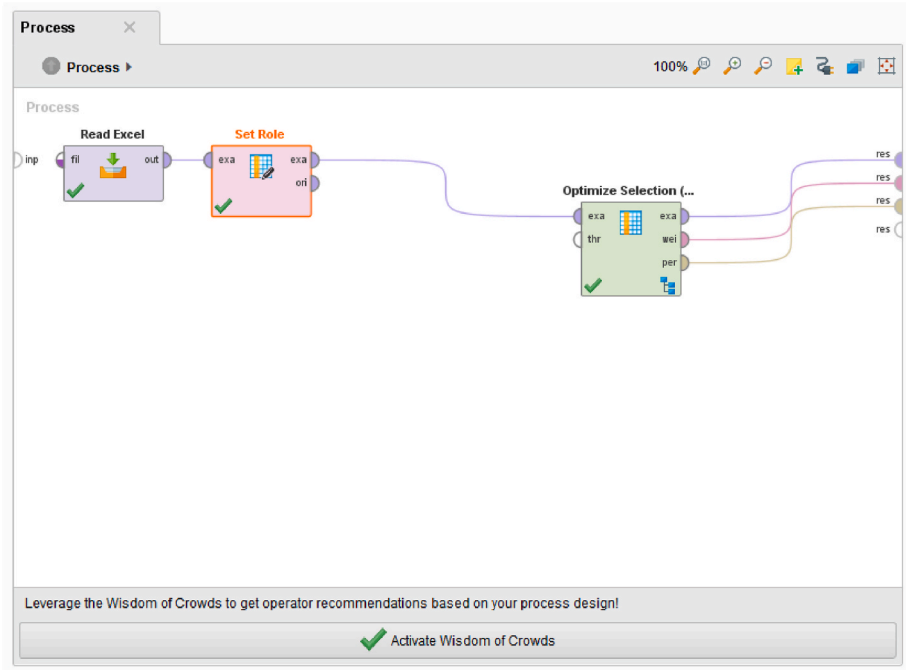

Fig. 2. Using the optimize selection module.

as it can generate interpretable "if-then" rules, which is valuable for providing actionable insights to healthcare professionals.

- K-Nearest Neighbors (KNN): KNN was chosen for its simplicity and effectiveness in classification problems where the decision boundary is not clearly linear. This algorithm is useful for heart attack prediction because it considers the proximity of similar patients, making it a strong choice when dealing with a variety of patient characteristics.
- Neural Networks: We included neural networks because of their ability to model complex, non-linear relationships in data. Given the complexity of heart disease risk factors, this algorithm is useful for capturing intricate patterns that other methods may miss, allowing for more nuanced predictions.
- Naive Bayes: The Naive Bayes algorithm was selected due to its probabilistic nature, which is effective in scenarios where independent variables are assumed to influence the outcome independently. It is particularly useful when we have categorical data, as is the case with some of the patient characteristics in this study.
- Support Vector Machine (SVM): SVM was included for its effectiveness in high-dimensional spaces and its ability to handle both linear and non-linear classification problems. While SVM showed a lower accuracy in this study, it was still considered valuable for comparison with other algorithms to assess the robustness of the predictive model.
- Logistic Regression: Logistic regression was selected for its simplicity and interpretability in binary classification tasks. Given the binary nature of the dependent variable (heart attack: yes/no), logistic regression provides clear, probabilistic outputs that were useful for assessing risk.

Fig. 3, displays the modules employed in decision tree algorithm. When result of this algorithm was juxtaposed with the actual data, it came to light that out of the 38 patients, the decision tree accurately predicted 37 instances, signifying a predictive accuracy of 97 percent for this method.

The modules employed in neural network algorithm are depicted in Fig. 4. Upon comparing result of the neural network algorithm with the actual data, it was noted that out of the 38 patients, the neural network correctly predicted 32 instances, resulting in a predictive accuracy of 84% for this method.

The modules utilized in Naive Bayes algorithm are displayed in Fig. 5. After comparing result of the Naive Bayes algorithm with the actual data, it was discovered that out of the 38 patients, Naive Bayes accurately predicted 29 instances, yielding a predictive accuracy of 76%

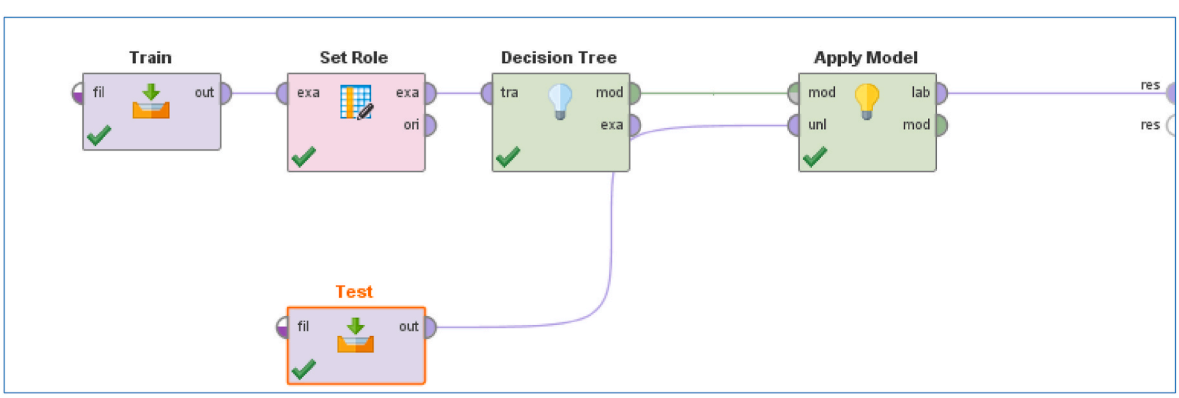

Fig. 3. Decision tree algorithm.

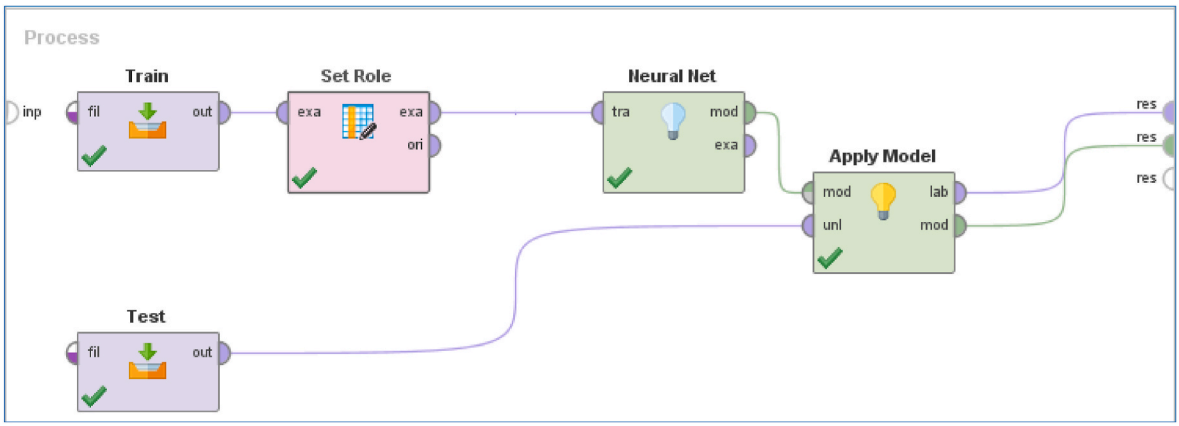

Fig. 4. Neural network algorithm.

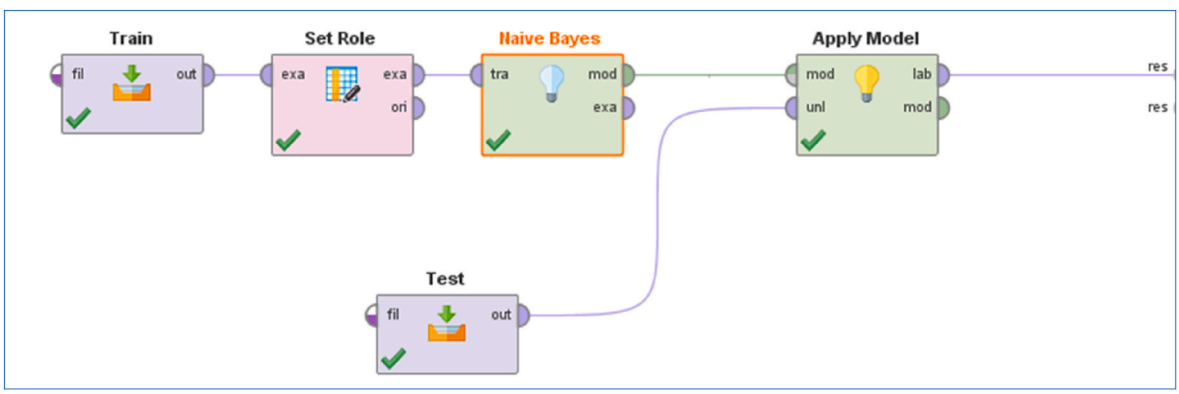

Fig. 5. Naive Bayes algorithm.

for this method.

The modules employed in K-Nearest Neighbors (KNN) algorithm are exhibited in Fig. 6. After comparing result of the KNN algorithm with the actual data, it was noticed that out of the 38 patients, KNN accurately predicted 34 instances, resulting in a predictive accuracy of 89% for this method.

Fig. 7 displays the modules utilized in Support Vector Machine (SVM) algorithm. Upon comparing result of the SVM algorithm with the actual data, it was noted that out of the 38 patients, SVM accurately predicted 26 instances, resulting in a predictive accuracy of 68% for this method.

The modules employed in Logistic Regression algorithm are illustrated in Fig. 8. After contrasting result of the Logistic Regression algorithm with the actual data, it was discovered that out of the 38 patients, the Logistic Regression algorithm accurately predicted 31 instances, resulting in a predictive accuracy of 82% for this method.

The primary metric used to assess the performance of each algorithm was prediction accuracy, which measures the proportion of correctly classified instances (heart attack occurrence vs. non-occurrence) out of all instances in the test set. This was calculated by comparing the predicted outcomes to the actual outcomes for each patient in the testing data. By using this criterion, we ensure a thorough evaluation of the algorithms and gain a deeper understanding of their strengths and limitations in predicting heart attack risk. Table 4 displays the predictive accuracy of various algorithms for the recurrence of heart disease. As indicated in the table, decision tree algorithm attained maximum accuracy of 97% compared to all other algorithms (see Table 5).

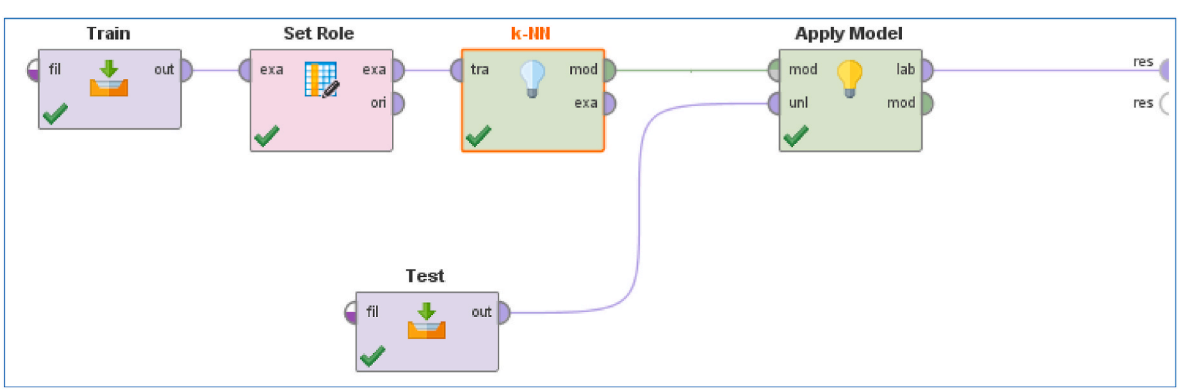

Fig. 6. K-nearest neighbors algorithm.

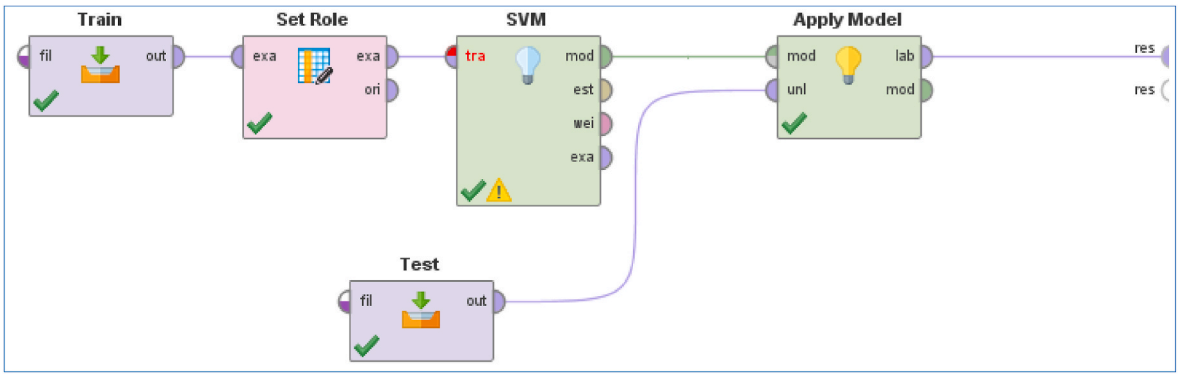

Fig. 7. Support vector machine algorithm.

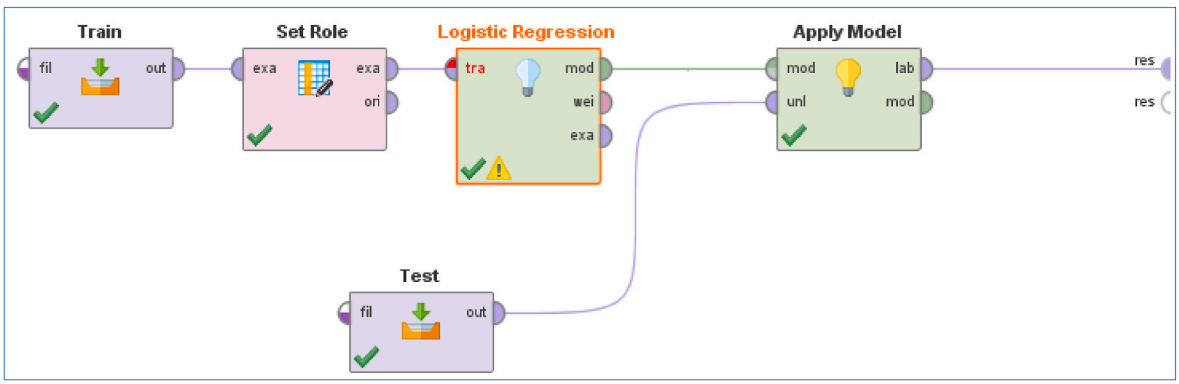

Fig. 8. Logistic regression algorithm.

Table 4
The key factors affecting heart attack.

Row	Independent Variable (Predictive Factors)
1	age
2	chest pain type
3	diabetes
4	lifestyle
5	Resting-electrocardiographic-results
6	smoking
7	Trait_Anxiety

Table 5
The predictive accuracy.

Algorithm	Predictive Accuracy
Decision Tree	97%
KNN	89%
Neural Network	84%
Logistic Regressin	82%
Naive Bayes	76%
SVM	68%

4.3. Analysis of machine learning results based on decision tree algorithm

Table 4 demonstrates that the decision tree algorithm is the best algorithm based on the data of this research for predicting heart attack. Consequently, the software produced the following graph based on the decision tree algorithm:

Fig. 9 displays the results of the data analysis performed on all collected data. Based on the analysis of the algorithm results, the following rules were extracted:

1 Patients with a trait_anxiety greater than 67.5 but without or with mild chest pain are highly likely to suffer from a heart attack.
2 Patients with a trait_anxiety greater than 42.5 with severe chest pain are highly likely to suffer from a heart attack.
3 Patients who smoke and have severe chest pain at the age of over 58.5 are likely to suffer from a heart attack.
4 Patients with unbearable chest pain and unsatisfactory resting-electrocardiographic-results at the age of over 68 are highly likely to experience a heart attack.

Out of the seven factors identified as impacting the likelihood of heart attack in Table 3, the decision tree rules highlight five of them as particularly significant. These five factors are chest pain type, trait_anxiety, age, smoking, and resting-electrocardiographic-results. Prioritizing the identification and management of these influential factors could have a significant impact on prediction of heart attack while diabetes and lifestyle factors play a lesser role. Therefore, it is recommended that physicians give special attention to these five factors. If the rules are examined more carefully, it can be observed that two factors among the five selected in decision tree rules have more emphasis: chest pain, trait_anxiety. The combination of these two factors is crucial in determining the likelihood of a heart attack. As depicted in Fig. 9, individuals experiencing intense chest pain or high levels of trait_anxiety are at an increased risk of suffering from a heart attack.

Nonetheless, further examination is necessary to validate the importance of these influential variables. The question in this regard is whether the five above factors predict reality without considering other factors? In order to address this query, it becomes essential for evaluating the validity of the identified influential variables based on the extracted rules. One suitable method for validating these rules is to use fuzzy clustering. Since clustering does not specify any categories or criteria for data separation, it can be used as an indicator of the accuracy

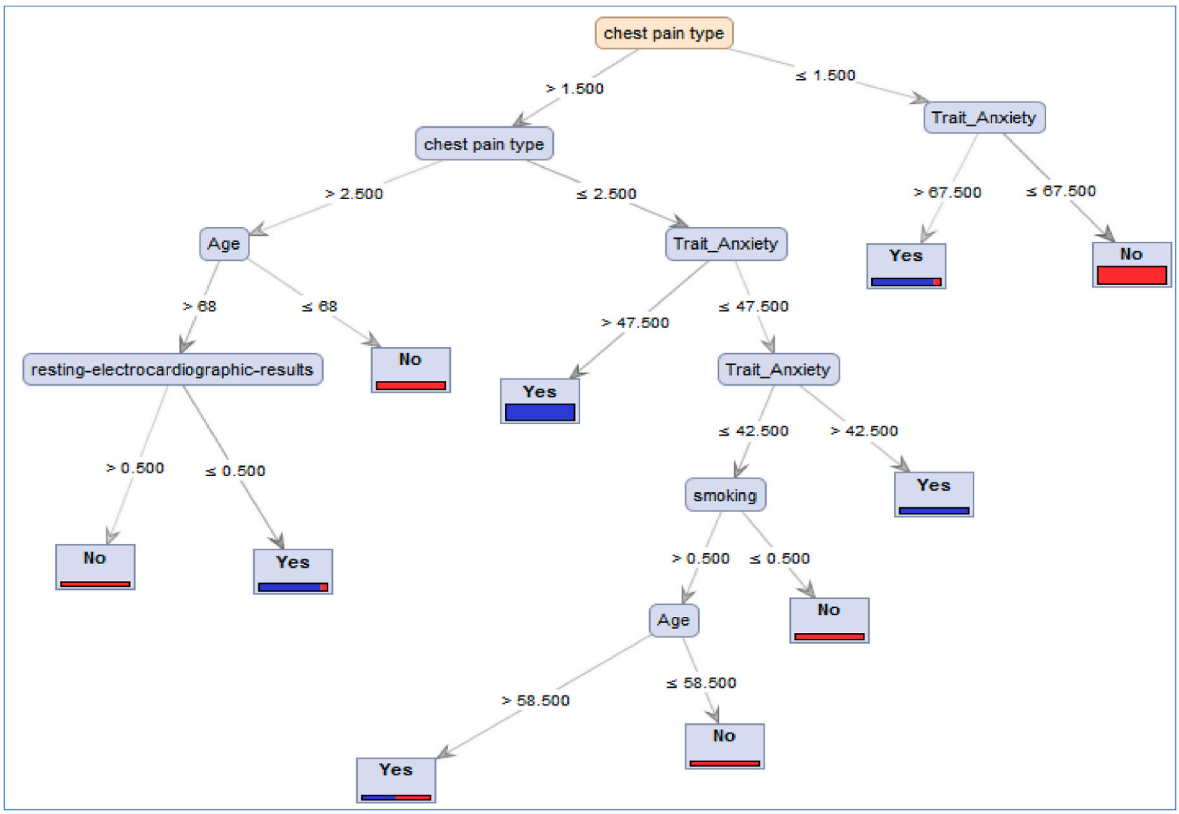

Fig. 9. Decision tree graph.

of the rules extracted from the decision tree by comparing the segmentation obtained from clustering with the actual data. The primary advantage of fuzzy clustering is the division of data into multiple clusters, with each data point being allocated a degree of membership to each cluster. It allows for greater adaptability in representing the likelihood that data points could belong to multiple clusters.

4.4. Using fuzzy clustering

In the previous section, based on the rules extracted from the decision tree, it was determined that among the 7 attributes, 5 affect the determination of branches.

Fig. 10 shows the modules used in the fuzzy clustering:

Based on frequency of the five selected influential attributes in the obtained rules, chest pain and trait_ anxiety are prioritized as the highest priority attributes, followed by smoking, age, and resting-electrocardiographic-results. It is currently looking for proof of the influencing variables in a heart attack by changing the variables in the implementation of the fuzzy clustering. As such, it is considering the following scenarios:

4.4.1. Scenario 1 - the implementation of fuzzy clustering with one attribute: chest pain

Based on the comparison of 138 real data related to heart attack and the results of fuzzy clustering in scenario 1, an alignment rate of 63% is observed.

4.4.2. Scenario 2 - the implementation of fuzzy clustering with one attribute: trait anxiety

Based on the comparison of 138 real data related to heart attack and the results of fuzzy clustering in scenario 2, an alignment rate of 61% is observed.

4.4.3. Scenario 3 - the implementation of fuzzy clustering with two attributes: chest pain and trait anxiety

After comparing 138 real heart attack-related data points with the

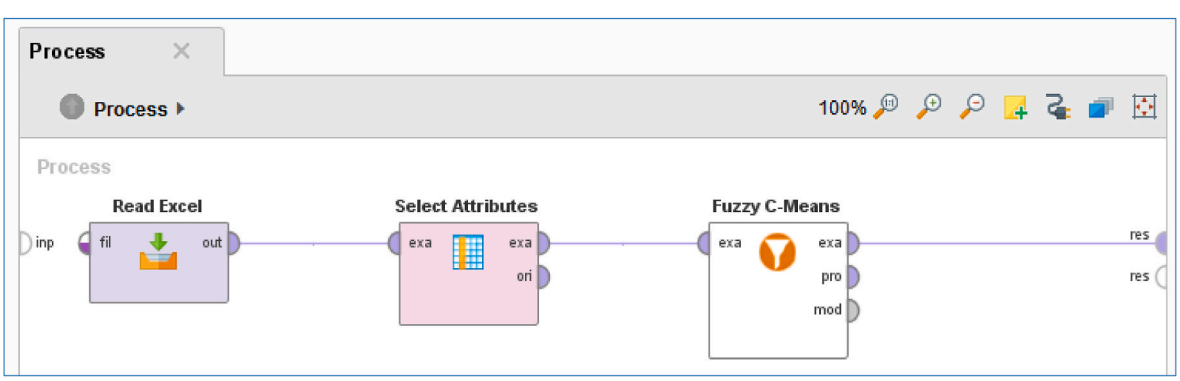

Fig. 10. Fuzzy clustering.

results of fuzzy clustering in scenario 3, an alignment rate of 76% was observed, indicating an acceptable level of accuracy in clustering. Consequently, it can be concluded that both chest pain and trait_anxiety factors, when considered together, impact the fuzzy clustering 's accuracy and can be given high priority in this analysis.

4.4.4. Scenario 4 - the implementation of fuzzy clustering with five attributes: chest pain, trait anxiety, age, smoking and resting-electrocardiographic-results

After analyzing actual data on heart attacks and applying fuzzy clustering in Scenario 4, an alignment rate of 82% was observed, indicating only minor differences between the third and fourth scenarios. Consequently, it can be concluded that the factors considered influential—age, smoking, and resting electrocardiographic results—have a lesser impact on clustering accuracy and should be given less priority in this analysis.

4.4.5. Scenario 5 - the implementation of fuzzy clustering with seven attributes: chest pain, trait anxiety, age, smoking, resting-electrocardiographic-results, diabetes and lifestyle

After analyzing actual data on heart attacks and applying fuzzy clustering in Scenario 5, an alignment rate of 82% was observed, indicating no significant difference between Scenarios 4 and 5. Consequently, it can be concluded that diabetes and lifestyle factors do not affect the accuracy of the fuzzy clustering and may not be considered influential factors in this analysis.

4.5. Analyzing

The data analysis techniques in this study followed a structured approach to ensure reliability and validity. Feature selection was performed using the "Optimize Selection" module in RapidMiner, which identified the seven most significant variables affecting heart attack risk out of 13, based on their predictive importance and validation by hospital physicians. A range of machine learning algorithms, including Decision Tree, Neural Network, K-Nearest Neighbors (KNN), and others, was evaluated for their strengths in classification problems. The dataset was divided into training (100 patients) and testing (35 patients) subsets to prevent overfitting, with models trained on the former and validated on the latter.

The rules generated by the Decision Tree algorithm were further assessed using Fuzzy C-Means clustering to verify the reliability of the identified influential variables across various scenarios, aligning predictions with actual outcomes. Accuracy served as the primary performance metric, supplemented by qualitative validation from hospital physicians. This comprehensive process ensured robust findings and established a clear framework for reproducibility.

After analyzing actual data on heart attack and applying fuzzy clustering in scenario 5, an alignment rate of 82% was observed, indicating a lack of difference between the fourth and fifth scenarios. Consequently, it can be concluded that diabetes and lifestyle factors do not impact the accuracy of the fuzzy C-means algorithm and may not be considered as influential factors in this analysis.

Table 6 provides a summary of the novel methodology.

The findings of this study bridge critical gaps in existing literature by emphasizing data-driven feature selection, enhancing model interpretability, and leveraging hybrid approaches. Unlike prior studies that rely on subjective expert opinions, this research employs objective methods like recursive feature elimination for robust feature selection. Additionally, it prioritizes interpretability alongside accuracy, addressing the need for transparent decision-making in healthcare. Lastly, the study integrates clustering techniques with predictive algorithms, advancing hybrid approaches to improve both performance and actionable insights. These contributions collectively enhance methodological rigor and practical applicability in healthcare settings.

Table 6
Summary of the novel methodology.

Step	Description
Feature Selection	Identification of the most critical heart attack predictors.
Rule Extraction	Deriving interpretable decision rules from machine learning.
Validation	Fuzzy clustering to verify predictive accuracy and reliability.

5. Conclusion

The objective of this research was to determine and select some factors that significantly contribute to heart attack among all possible factors. To achieve this, after data preprocessing, the feature selection module in the software was utilized. Specifically, the optimize selection module in Rapid Miner software was employed to select the crucial features of a heart attack from 13 independent variables gathered. Interestingly, only seven variables were selected from the 13 independent variables in the patients' data. The software assessed the accuracy rate of estimating the variable of a heart attack using the chosen variables, which was found to be 97.86%. Notably, the software's output accuracy was confirmed by hospital physicians. The findings of this study address the research objectives comprehensively:

- Identifying Influential Factors: By isolating seven significant predictors, the research contributes to the understanding of key variables associated with heart attacks, enabling targeted interventions in clinical settings.
- Predictive Accuracy: The superior performance of the decision tree algorithm validates the effectiveness of machine learning in achieving accurate predictions, meeting the objective of developing reliable predictive models.
- Validation Through Clustering: The high alignment rates observed with fuzzy clustering provide additional evidence for the robustness of the decision tree rules, aligning with the objective of validating the identified influential factors.

Based on the decision tree rules, key factors were identified for predicting the likelihood of a heart attack. Finally, Fuzzy clustering was used to confirm the validity of the rules. Our use of fuzzy clustering to validate the decision tree results provides a novel approach in literature, as it allows for a more nuanced understanding of the impact of each variable. Our clustering results confirmed that chest pain and trait anxiety were the most influential variables in predicting heart attack, which suggests that these factors should be prioritized in clinical assessments. Previous studies often relied solely on predictive accuracy without validating the importance of the features selected by machine learning models, making our validation approach a significant contribution to the field.

In this study, some areas were improved, but previous research works were inadequacies:

- Implementing the Feature Selection Module within the Software: In previous studies, feature selection was only based on expert opinions, but in this research, feature selection was performed based on software and confirmation by physicians.
- Extracting If … Then … Rules from the Decision Tree Algorithm's Output: Previous research has mainly focused on selecting the algorithm that provides better predictions, but in this research, in addition to prediction, the aim is to discover knowledge that can greatly assist physicians in making more accurate and faster decisions.
- Validation of the Extracted Rules: By using fuzzy clustering, the validity of the approved rules was confirmed, and this method was proposed as an innovative approach rather than previous studies.

5.1. Future studies

This study provides valuable insights into predicting heart attacks using machine learning algorithms, but it has several limitations. The small sample size of 135 patients limits the generalizability of the findings, and future research should include larger, multicenter datasets. Additionally, relying on data from a single hospital introduces potential biases tied to that institution's demographics and practices, which could be mitigated by expanding data collection to multiple hospitals and regions. The retrospective design of the study also poses risks of incomplete records and unaccounted confounding variables, suggesting that prospective studies could provide more consistent and complete data. Despite these challenges, the study remains a useful exploratory investigation, and addressing these limitations in future research will enhance the findings' practical value. While the study primarily employed a data-driven approach to identify key predictors of heart attack risk, it did not explicitly address the inclusion of control variables in the analysis. Future studies should aim to incorporate such variables.

Building upon the findings of this research, several specific areas for future investigation have been identified to further enhance the understanding and practical application of predictive models for heart attack risk:

- Expanding Dataset Size and Diversity: Future studies should consider incorporating larger datasets with more diverse patient demographics, such as different ethnicities, socioeconomic backgrounds, and geographical locations. This would improve the generalizability of the findings and ensure the models remain robust across various populations.
- Incorporating Additional Risk Factors: While this study focused on 13 independent variables (predictive factors), future research could integrate other potential risk factors, such as genetic predisposition, long-term medication use, and environmental influences. This expansion could lead to a more comprehensive model for predicting heart attack risks.
- Advanced Fuzzy Classification Methods: As the fuzzy clustering technique in this study validated the extracted rules, future research could explore advanced fuzzy classification algorithms, such as Fuzzy Rough Sets or Neuro-Fuzzy Systems, to compare their effectiveness with traditional machine learning models.
- Developing and Testing an Expert System: Based on the findings, future studies could focus on designing and testing an expert system for real-time risk prediction and patient management in hospital admission units. Such a system could provide actionable insights and recommendations to healthcare professionals, enabling timely interventions for high-risk patients.
- Comparative Analysis with Other Methods: Future research could compare the performance of the current predictive models with alternative approaches, such as deep learning techniques or ensemble methods, to evaluate their potential for further accuracy improvements.

By addressing these areas, subsequent research can build on the foundational work presented in this study and contribute to advancing predictive analytics. While machine learning approaches have shown significant promise in predicting heart attack risk, their limitations must be acknowledged. Overfitting, challenges in model interpretability, and dependency on data quality highlight areas for improvement. These considerations should guide future efforts to refine the application of machine learning in clinical decision-making.

CRediT authorship contribution statement

Mohammad Dehghani Saryazdi: Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali Mostafaeipour:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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